

Wireless Networks

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Wireless Networks

Lecture-2



Contents

- ☑ Power Units
- ☑ Coding Terminologies
- ☑ Modulation
- ☑ Coding
- ☑ Channel Capacity
- ☑ Wireless Access Channels
- ☑ Spread Spectrum



Decibels, dB

- ✓ **A decibel (dB)** is a logarithmic unit used to express the ratio between two values, typically power or intensity.
- ✓ It is commonly used in fields like **acoustics**, **electronics**, and **communications** to measure sound levels, signal strength, and gain or loss in a system.

✓ **Mathematically**, the decibel is calculated as:

- For power ratio

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$

where: P_1 and P_2 are initial and final power levels.

- For voltage or current ratio

$$dB = 10 \log_{10} \frac{V_2}{V_1}$$

where: V_1 and V_2 are initial and final voltage/current.

Decibels, dB

☑ Why Do We Use Decibels?

- 1) Handles Large Ranges Easily.
- 2) Simplifies Multiplication and Division.

☑ Examples of Decibels in Use

▪ Communication and Electronics:

1. Wi-Fi signal strength: **-30 dBm** (strong) to **-90 dBm** (weak)
2. Amplifier gain: **+10 dB** means the output power is **10×** the input power
3. Attenuation in fiber optics: **-3 dB** means power is reduced by **50%**

▪ Radio and Telecommunications:

1. Mobile network signal: **-50 dBm** (excellent) to **-120 dBm** (poor)
2. Decibel milliwatts (dBm) for measuring radio signals: **0 dBm = 1 mW**

Decibels, dB

Large Power Ratios (Amplification)

Power Ratio (P_2/P_1)	dB Value
1 (same power)	0 dB
10× increase	10 dB
100× increase	20 dB
1,000× increase	30 dB
10,000× increase	40 dB
1,000,000× increase	60 dB

For example, if an amplifier boosts a signal from 1 mW to 1 W (**1,000× increase**), it can be simply written as **30 dB gain** instead of saying "1,000 times more power."

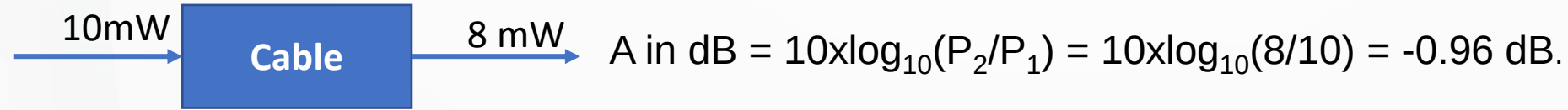
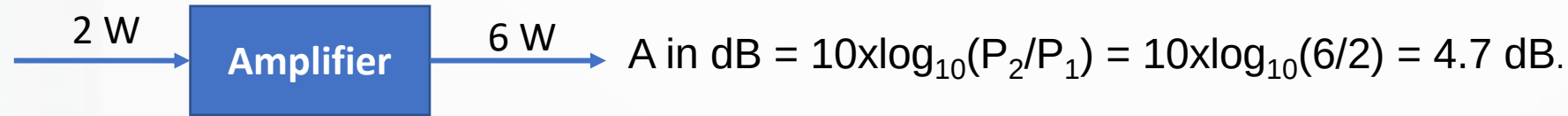
Decibels, dB

Small Power Ratios (Attenuation)

Power Ratio (P_2/P_1)	dB Value
1 (same power)	0 dB
1/10 (10% of original)	-10 dB
1/100 (1% of original)	-20 dB
1/1,000 (0.1% of original)	-30 dB
1/10,000 (0.01% of original)	-40 dB
1/1,000,000 (0.0001% of original)	-60 dB

For example, if a fiber optic cable reduces a signal's power to **1/1,000** of its original strength, it can be described as **-30 dB loss**, making it easier to express than saying "the power is now 0.1% of the original."

Decibels



If the final result is **positive** then A is **Gain**. If the final result is **negative**, then A is **loss**.

☑ In the following Figure, calculate the power of the received signal.



$$3 = 10 \times \log_{10}(P_2/1) \rightarrow 3/10 = \log_{10}(P_2) \rightarrow P_2 = 10^{0.3} = 1.995 \text{ mW}$$

Decibels

▪ dB Laws

➤ Law of Zero :

- 0 dB when $P_2 = P_1$.

➤ Law of 3s:

- — 3 dB when $P_2 = 2P_1$
- -3 dB when $P_2 = 0.5P_1$.

➤ Law of 10s:

- 10 dB when $P_2 = 10P_1$
- -10 dB when $P_2 = 1/10 P_1$.

Key Topic

Table 1-3 Power Changes and Their Corresponding dB Values

Power Change	dB Value
=	0 dB
$\times 2$	+3 dB
$/ 2$	-3 dB
$\times 10$	+10 dB
$/ 10$	-10 dB

Decibels, dBm and dBW

☑ dBm indicates that the specified power is measured relative to 1 milliwatt. The mean your reference is 1 milliwatt

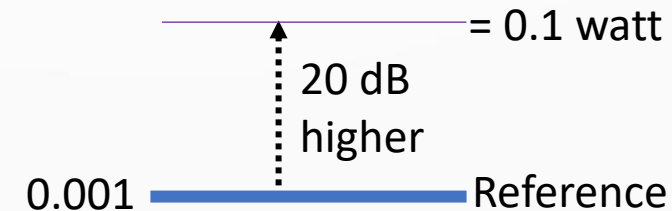
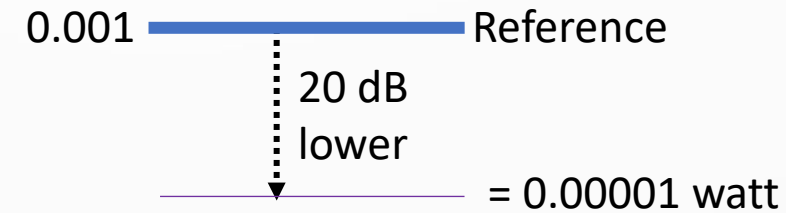
☑ -20 dBm does not indicate **negative power**; rather, it signifies that the power level is 20 dB lower than the reference level of 1 milliwatt = 0.001 watt.

☑ 20 dBm signifies that the power level is 20 dB higher than the reference level of 1 milliwatt = 0.001 watt.

☑ dBW indicates that the specified power is measured relative to 1 watt. The mean that your reference is 1 watt



$$\text{dBm} = 10 \log_{10} \frac{P_2}{0.001 \text{ W}}$$



$$\text{dBW} = 10 \log_{10} \frac{P_2}{1 \text{ W}}$$

Decibels

☑ Relation between dBm and dB

$$P = 20\text{dBm} - 3\text{ dB} = ?$$

$$P = 10\log_{10} \frac{0.1}{0.001} + 10\log_{10} \left(\frac{1}{2}\right)$$

- 20 dBm signifies that the power is increased by 100 times the reference value 0.001.
- -3 dB signifies that the power is reduced to half

$$\begin{aligned} P &= 10\log_{10} \frac{0.1 \times \frac{1}{2}}{0.001} \\ &= 17\text{ dBm} \end{aligned}$$

- The final result is in dBm because it is evaluated with reference to 1 milliwatt.

Decibels: Examples

Example 1

What is the attenuation in dB if the power is reduced by half (50% loss)?

Solution

$$\begin{aligned}\text{Attenuation in dB} &= 10\log_{10} (P_2/P_1) = 10\log_{10} (0.5 P_1 / P_1) = 10\log_{10}(1/2) \\ &= 10\log_{10}(1) - 10\log_{10}(2) = 0 - 3 \text{ dB} = -3 \text{ dB}\end{aligned}$$

Example 2

Compute the loss in dB if the received power of a 100 mW transmitted signal is only 1 mW

Solution

$$\begin{aligned}\text{Attenuation in dB} &= 10\log_{10} (P_2/P_1) = 10\log_{10} (1/100) \\ &= 10\log_{10}(0.01) = 10 \times -2 = -20 \text{ dB}\end{aligned}$$

Decibels: Examples

Example 5

Convert 1 Watt to dBm.

Solution

We have $10\log_{10}(1 \text{ W}/1 \text{ mW}) = 10\log_{10}(1000 \text{ mW}/1 \text{ mW}) = 30 \text{ dBm}$

Example 6

Express 1 mW in units of dBm

Solution

$10\log_{10}(1) = 10 \times 0 = 0 \text{ dBm}$ (ZERO dBm does not mean there is no power!)

Decibels: Examples

Example 7

With 100 μW of noise, what would be the SNR in dB if the received signal strength is 1 mW?

Solution

$P_{\text{signal}} = 1 \text{ mW}$ (received signal strength), $P_{\text{noise}} = 100 \mu\text{W}$

$$\text{SNR} = 10\log_{10}(1000/100) = 10\log_{10}(10) = 10 \text{ dB}$$

Example 8

Received signal strength is measured at 10 mW. What is the noise power if $\text{SNR} = 10 \text{ dB}$?

Solution

$$\text{SNR} = 10\text{dB} = 10\log_{10}(10 \text{ mW}/P_{\text{noise}})$$

$$P_{\text{noise}} = 1 \text{ mW}$$

Coding Terminologies

❖ Symbol:

- ✓ Smallest detectable **element of a signal** with defined amplitude, frequency, and phase.
- ✓ **Shorter** symbol duration allows for higher transmission of signal elements carrying bits per second.

❖ Baud Rate:

- ✓ Number of symbols transmitted per second.
- ✓ Inversely related to symbol duration; also known as symbol rate.
- ✓ Determines how quickly signal properties (amplitude, frequency, phase) can be modulated.

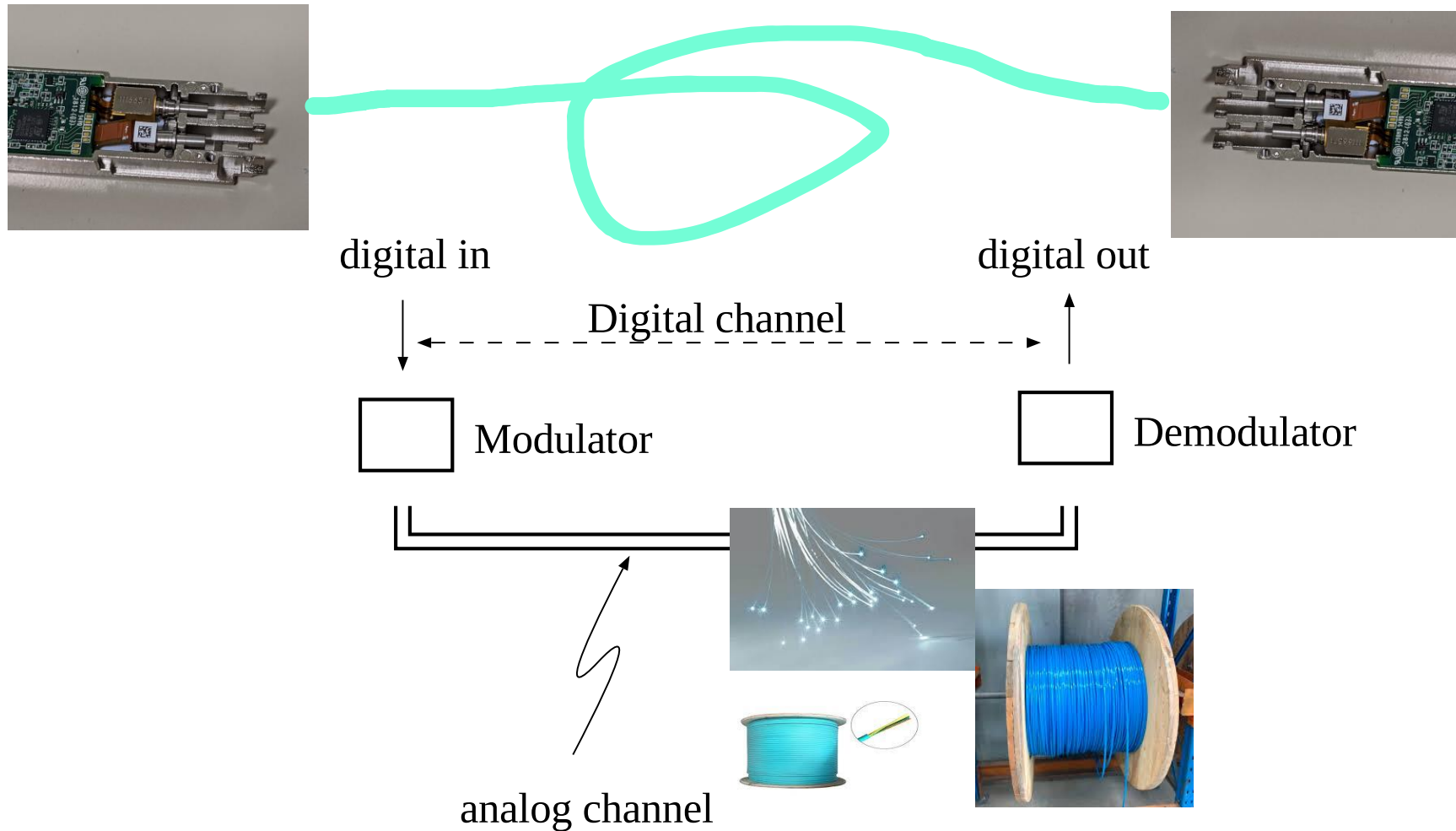
Coding Terminologies

❖ Data Rate:

- ✓ Measured in bits per second (bps).
- ✓ Represents the number of bits transmitted per second.
- ✓ For binary signals, **data rate = baud** rate since each **symbol carries 1 bit**.
- ✓ For M-ary signals, where M distinct symbols are used, data rate can be calculated as **$\log_2 M$** bits per symbol.

Modulation

- ☑ Modulation: transform an information signal into a another signal more suitable for transmission on a physical medium.

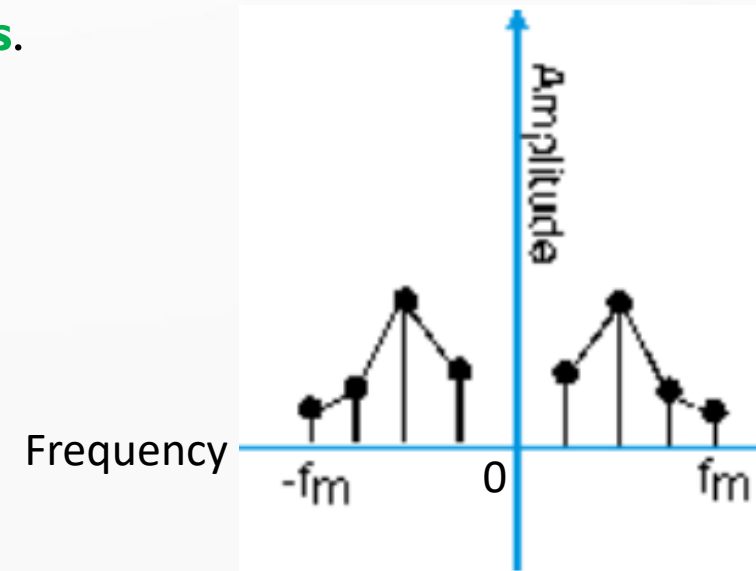


Modulation

Modulation: How the Process Begins

☑ Baseband signal

- An original, low-frequency signal carrying information such as audio, video, or data.
- Concentrated around 0 Hz and spans from $-f_m$ to $+f_m$. Audio signals (0 to 20Khz), video signals (0 to 55Khz)
- The spectrum typically spans from $-f_m$ to $+f_m$.
- All sources of information generate **baseband signals**.
- Typically used in **wired systems**.
- More **easily** affected **by low-frequency** noise.
- Suitable for **short distances**
- **Lower** power consumption



Modulation

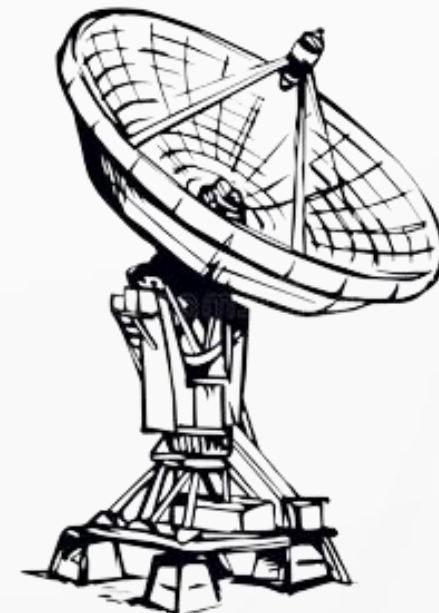
Modulation: How the Process Begins

- ☑ Let we want to send a **baseband** signal $x(t)$ from transmitter to receiver



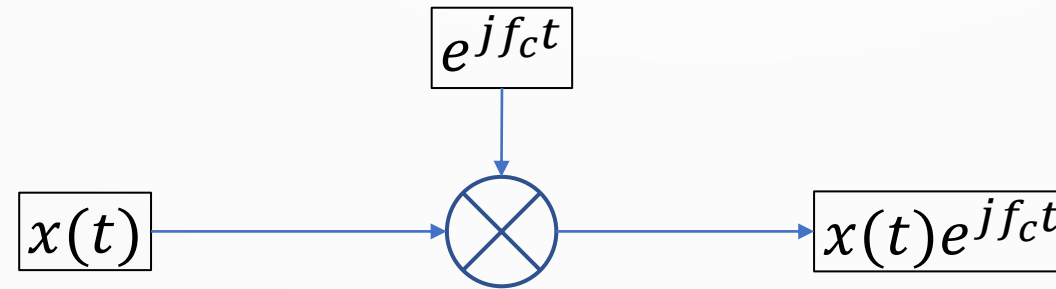
- ☑ The length of antenna $l \propto \frac{1}{f_m}$
- ☑ So, if f_m is small then the antenna must be large.
- ☑ In order to effectively receive the transmitted signal with low f_m , the length of the antenna and mobile antenna are so large.

**Instead of carrying small phone,
you must carry a giant phone cell.**

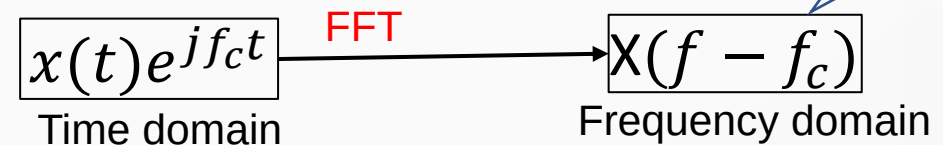


Modulation

- ✓ The modulation is used to make antenna size is feasible by **shifting** f_m to higher frequency.
- ✓ This shifting process can be executed by multiplying the signal $x(t)$ by the complex exponential function $e^{jf_c t}$ we get

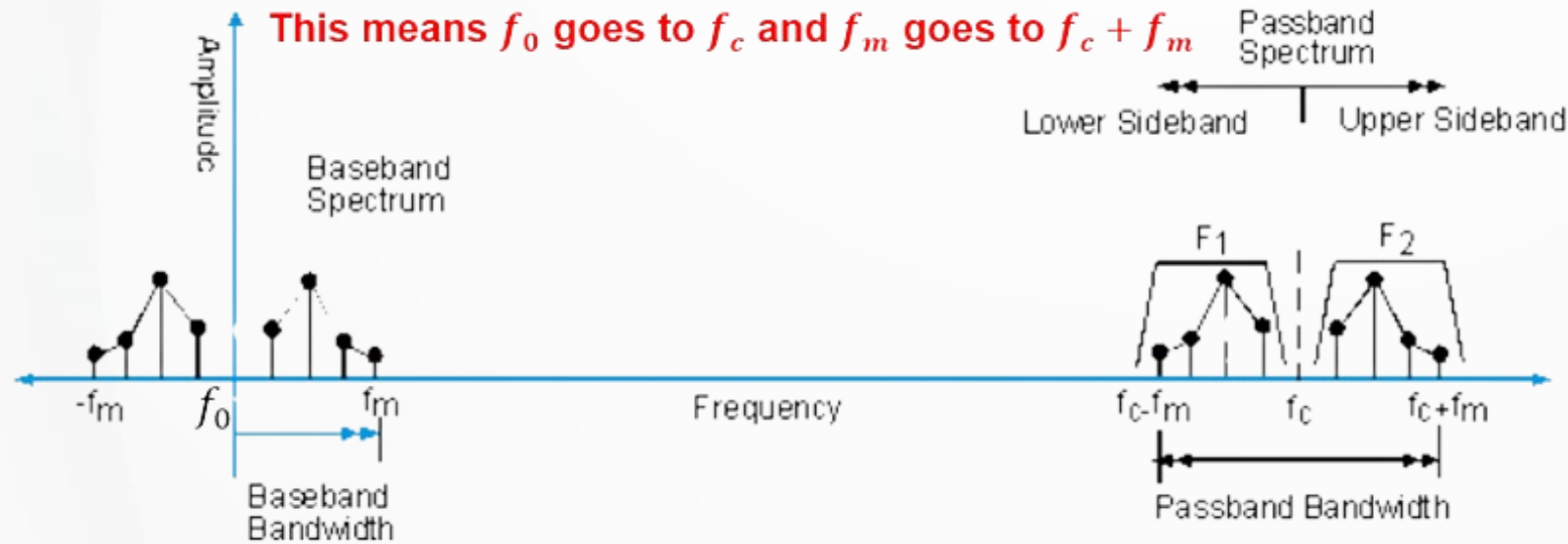


- ✓ From Fourier transform, we have



- ✓ $X(f)$ get shifted by f_c due to the multiplication of $x(t)$ by $e^{jf_c t}$.
- ✓ This is called frequency shifting property.

Modulation



✓ Why this process is useful ?

- If we transmit $x(t)$ in the form of $X(f - f_c)$ then the antenna length L is given as

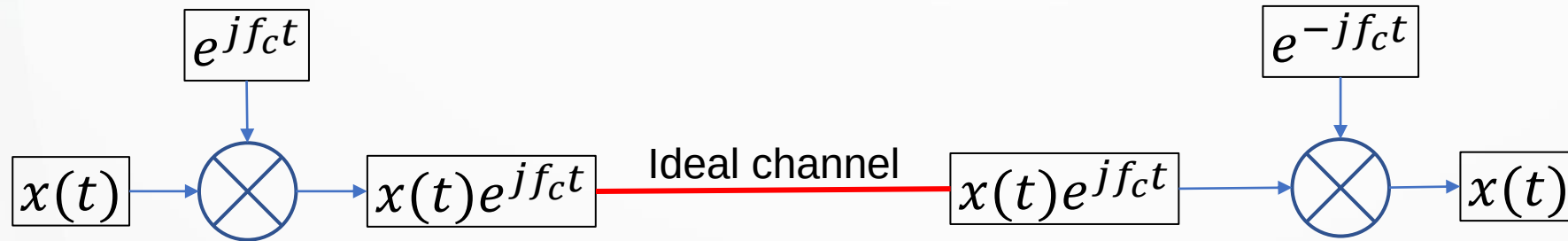
$$L \propto \frac{1}{f_c + f_m}$$

- If f_c is very large then L is very small.
- This is one of the main application of using modulation

Modulation

Demodulation: How the Process Begins

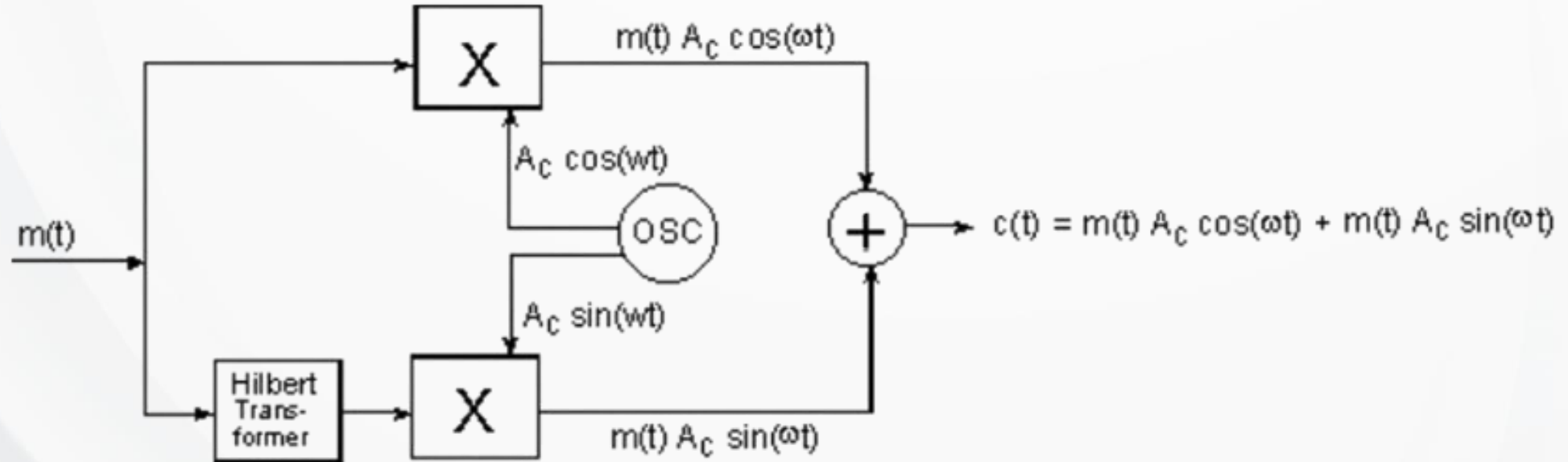
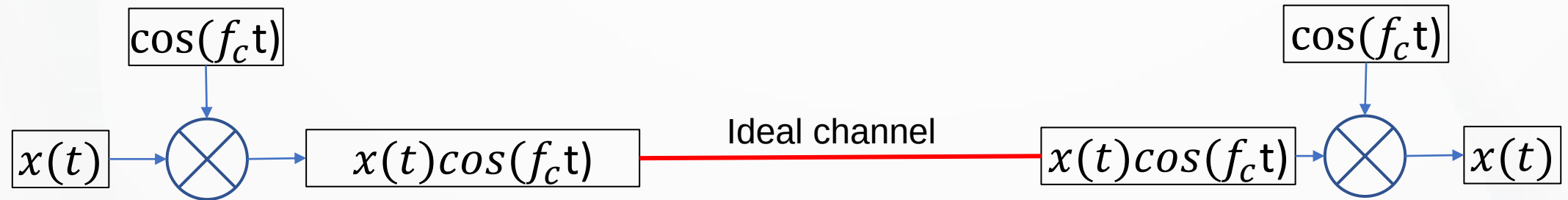
- At the receiver, we perform $x(t)e^{jf_c t} e^{-jf_c t}$ to extract $x(t)$
- This process is called demodulation



- The signal $e^{jf_c t}$ is called carrier signal.
- The frequency f_c is called carrier frequency which is usually very large number.
- In practical, we do not like to use complex numbers. That why we replace $e^{jf_c t}$ by **cos** and **sin** functions

Modulation

Modulation: How the Process Begins



Modulation

☑ Modulation Techniques

- ✓ **Amplitude Shift Keying (ASK):** Modulates amplitude of the carrier wave to represent 0's and 1's.

$$s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ 0 & \text{binary 0} \end{cases}$$

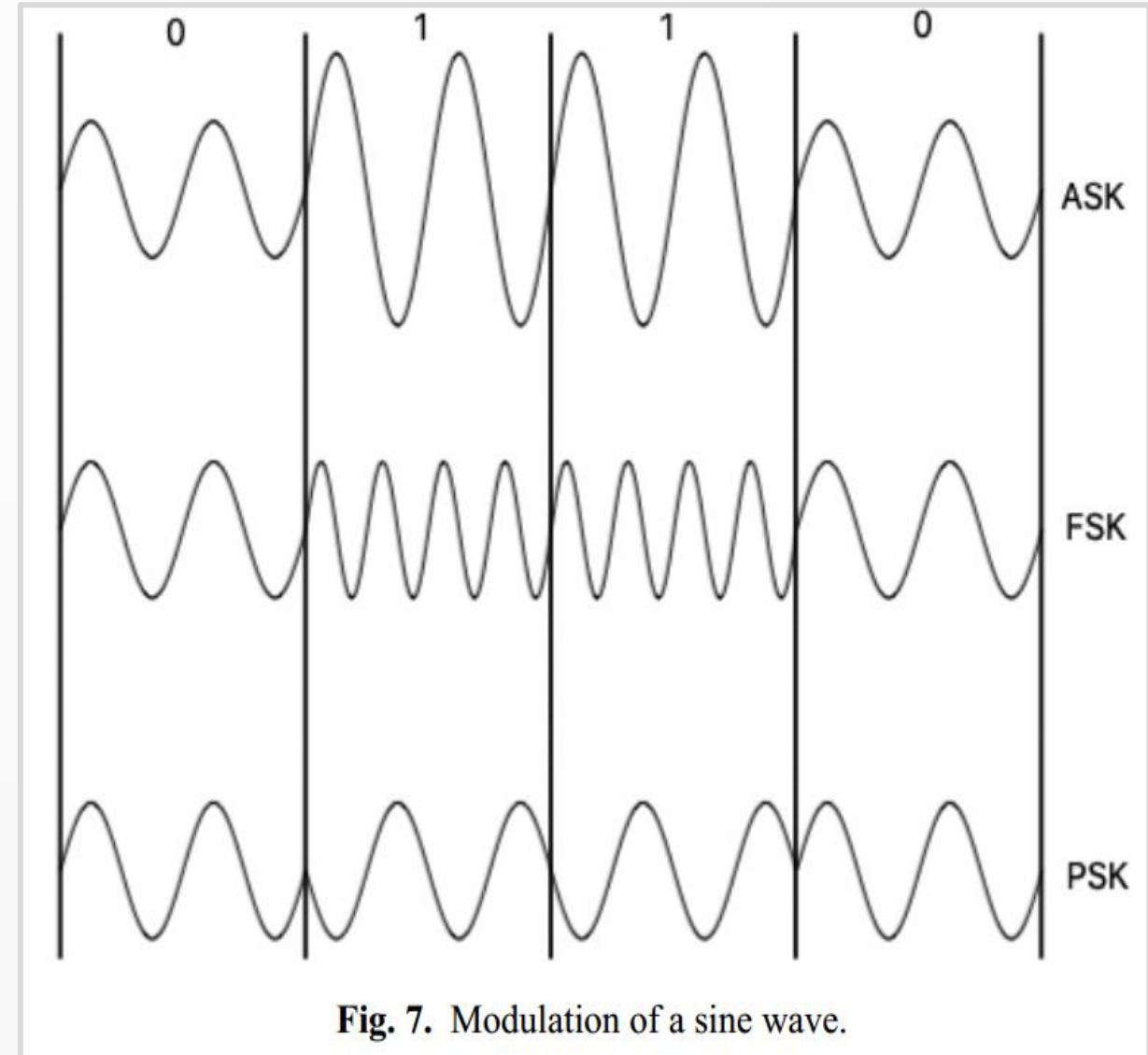


Fig. 7. Modulation of a sine wave.

Modulation

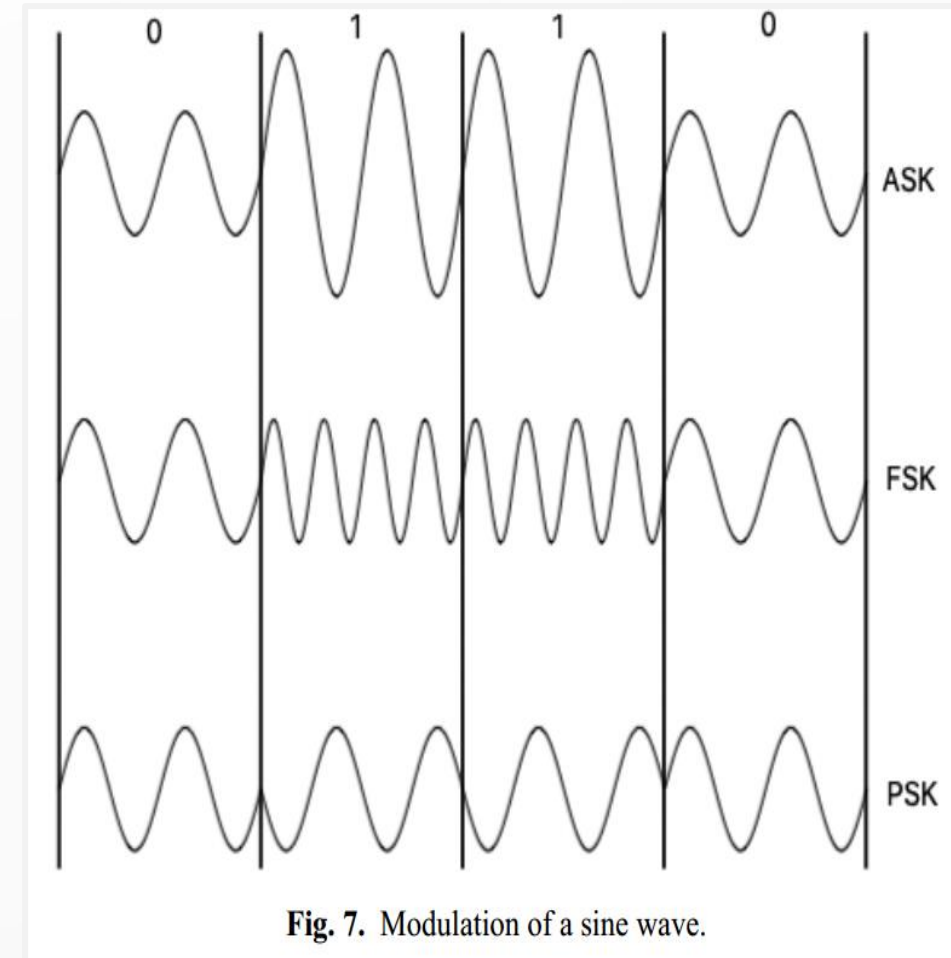
☑ Modulation Techniques

- ✓ **Frequency Shift Keying (FSK):** Modulates frequency of the carrier wave.

$$\text{BFSK} \quad B_s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary 1} \\ A \cos(2\pi f_2 t) & \text{binary 0} \end{cases}$$

- ✓ **Phase Shift Keying (PSK):** Modulates phase of the carrier wave.

$$\text{BPSK} \quad s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ A \cos(2\pi f_c t + \pi) & \text{binary 0} \end{cases} = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ -A \cos(2\pi f_c t) & \text{binary 0} \end{cases}$$



- ✓ These modulations maintain constant values of amplitude, frequency, or phase during fixed periods known as bit or symbol intervals.

Modulation

☑ Modulation Techniques

✓ Differential PSK (DPSK):

- In differential PSK (DPSK), phase changes indicate bits: no change for 0, change for 1.
- Simplifies receiver implementation by not requiring comparison against a reference phase.

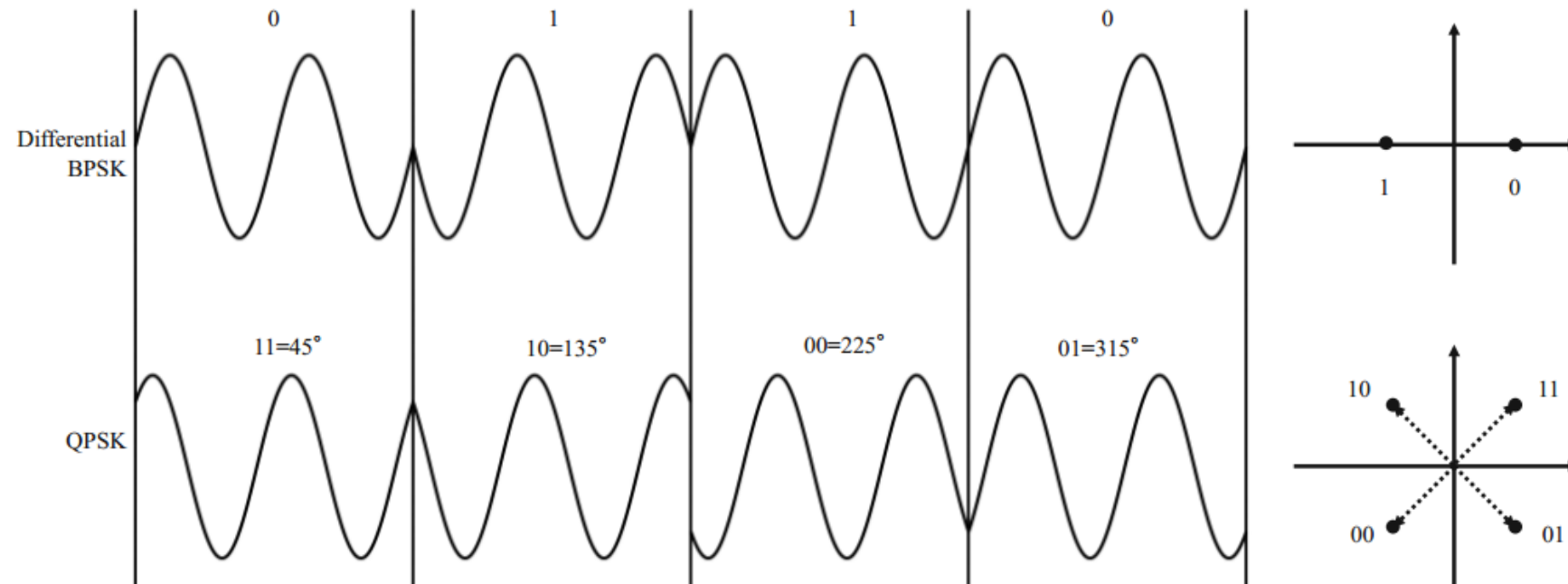


Fig. 8. Differential BPSK and QPSK.

Modulation

✓ Modulation Techniques

✓ BPSK (Binary PSK):

- Phase shift of 180° represents binary data (0 or 1).
- Shown in 2D (I-Q) graph with two distinct dots 180° apart.

✓ QPSK (Quadrature PSK):

- Uses four different phase shifts (0°, 90°, 180°, 270°).
- Each phase shift represents combinations of two bits per baud.
- 2D (I-Q) graph shows four distinct dots, each separated by 90 degrees.

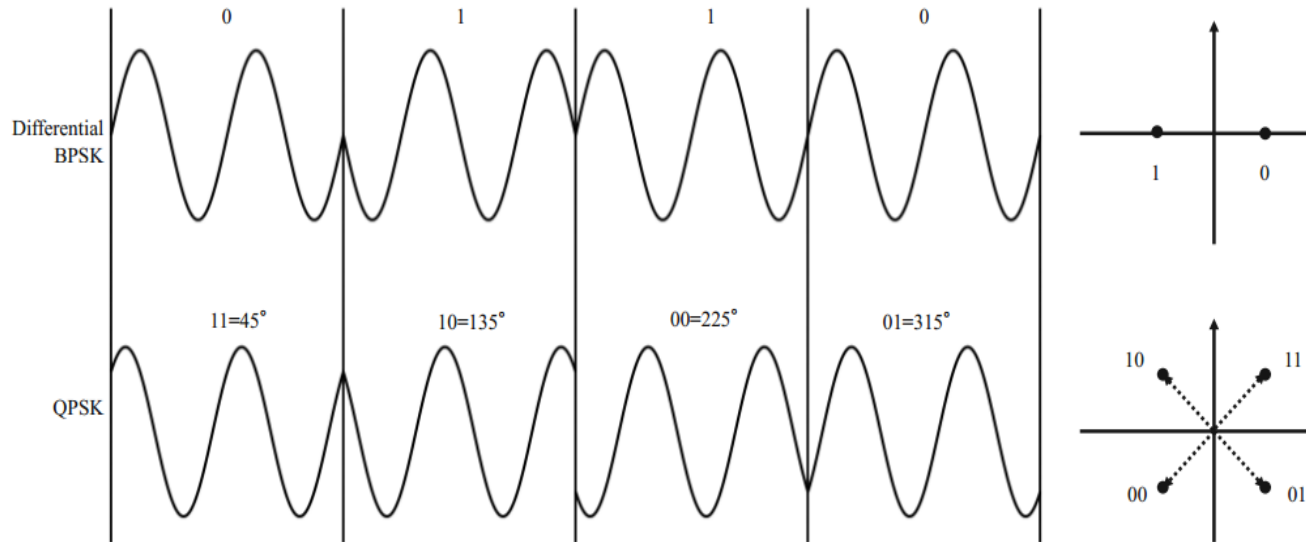


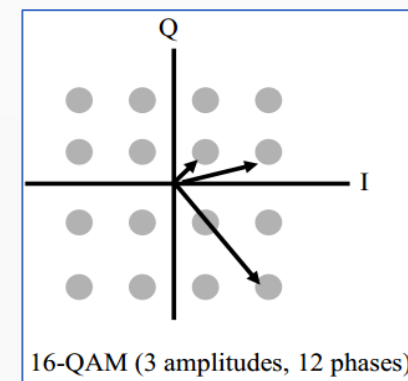
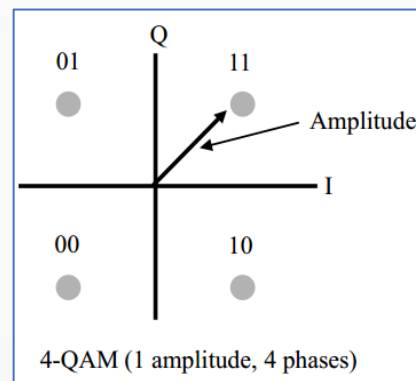
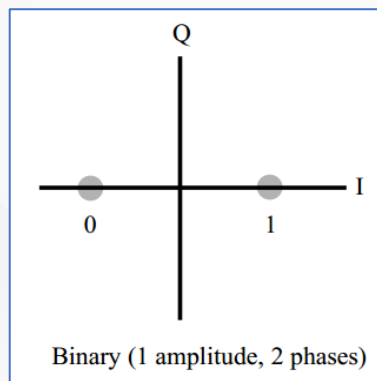
Fig. 8. Differential BPSK and QPSK.

$$s(t) = \begin{cases} A \cos \left(2\pi f_c t + \frac{\pi}{4} \right) & 11 \\ A \cos \left(2\pi f_c t + \frac{3\pi}{4} \right) & 01 \\ A \cos \left(2\pi f_c t - \frac{3\pi}{4} \right) & 00 \\ A \cos \left(2\pi f_c t - \frac{\pi}{4} \right) & 10 \end{cases}$$

Modulation

☑ Modulation Techniques

- **Quadrature Amplitude and Phase Modulation (QAM):**
 - ✓ Combines amplitude and phase modulations to increase data rate.
 - ✓ Achieves higher bit rates by using combinations of amplitude and phase.
- **Constellation Diagram:**
 - ✓ Visual representation of QAM signals.
 - ✓ Shows amplitude on one axis and phase on the other.



Modulation

☑ Modulation Techniques: **Types of QAM**

❖ **2-QAM (Binary Phase Shift Keying, BPSK):**

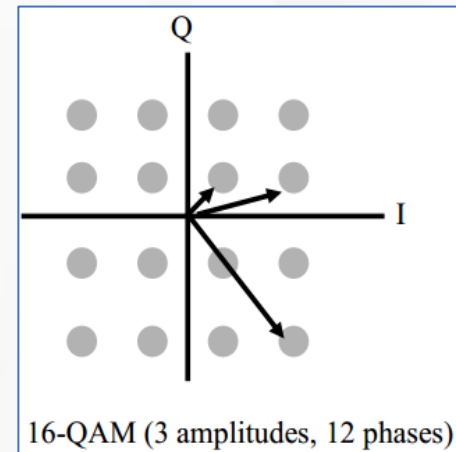
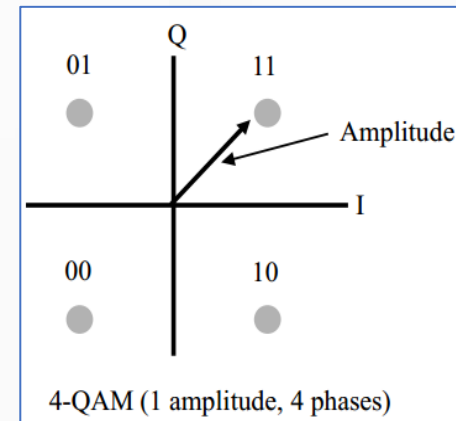
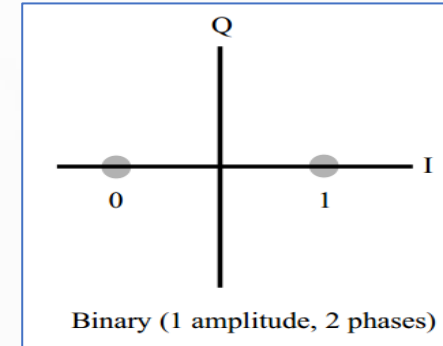
- Constant amplitude with 2 phase shifts (0° , 180°).
- Provides 1 bit per symbol.

❖ **4-QAM (Quadrature PSK, QPSK)**

- 4 phase shifts (0° , 90° , 180° , 270°) with 1 amplitude.
- Offers 2 bits per symbol.

❖ **16-QAM:**

- 3 different amplitudes combined with 4 phase shifts for each amplitude.
- Provides 16 combinations ($4 + 4 + 8$), allowing 4 bits per symbol.



Modulation

☑ Modulation Techniques: **Types of QAM**

❖ **Increasing Bit Rates:**

- ✓ Higher-order QAM (e.g., 64-QAM, 256-QAM, 1024-QAM) increase data rates by encoding more bits per symbol.

Wireless Technology	Supported QAM Technique
4G	256-QAM
5G	1024-QAM
WiFi 802.11n	16-QAM, 64-QAM
WiFi 802.11ac	256-QAM
WiFi 802.11ax	1024-QAM

❖ **Future Trends:**

- ✓ Latest wireless standards utilize high-order QAMs like 1024-QAM.
- ✓ Advancements in hardware and signal processing are expected to enable even higher QAMs in the future.

Coding – a channel function

- ☑ Coding means change the representation of data.

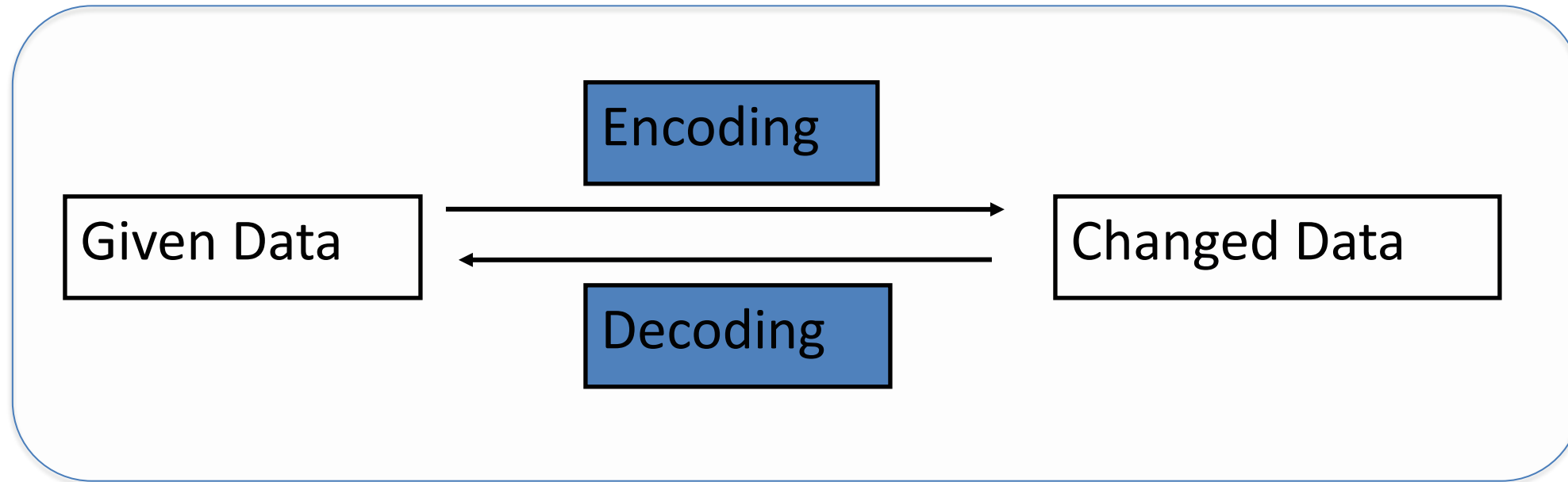
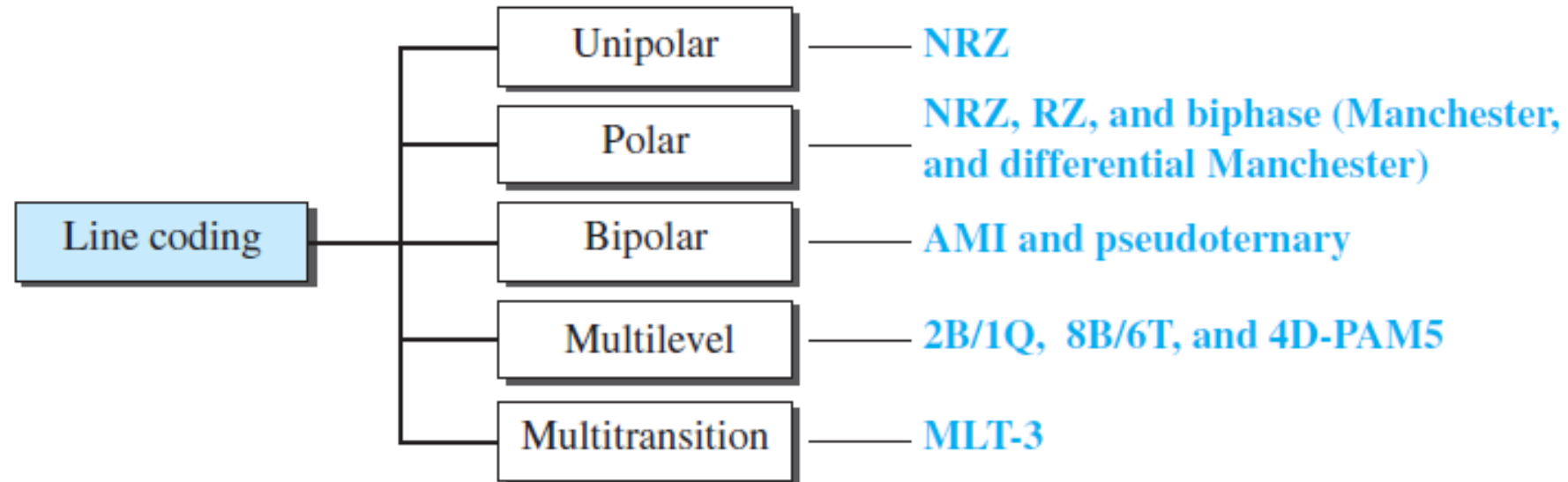
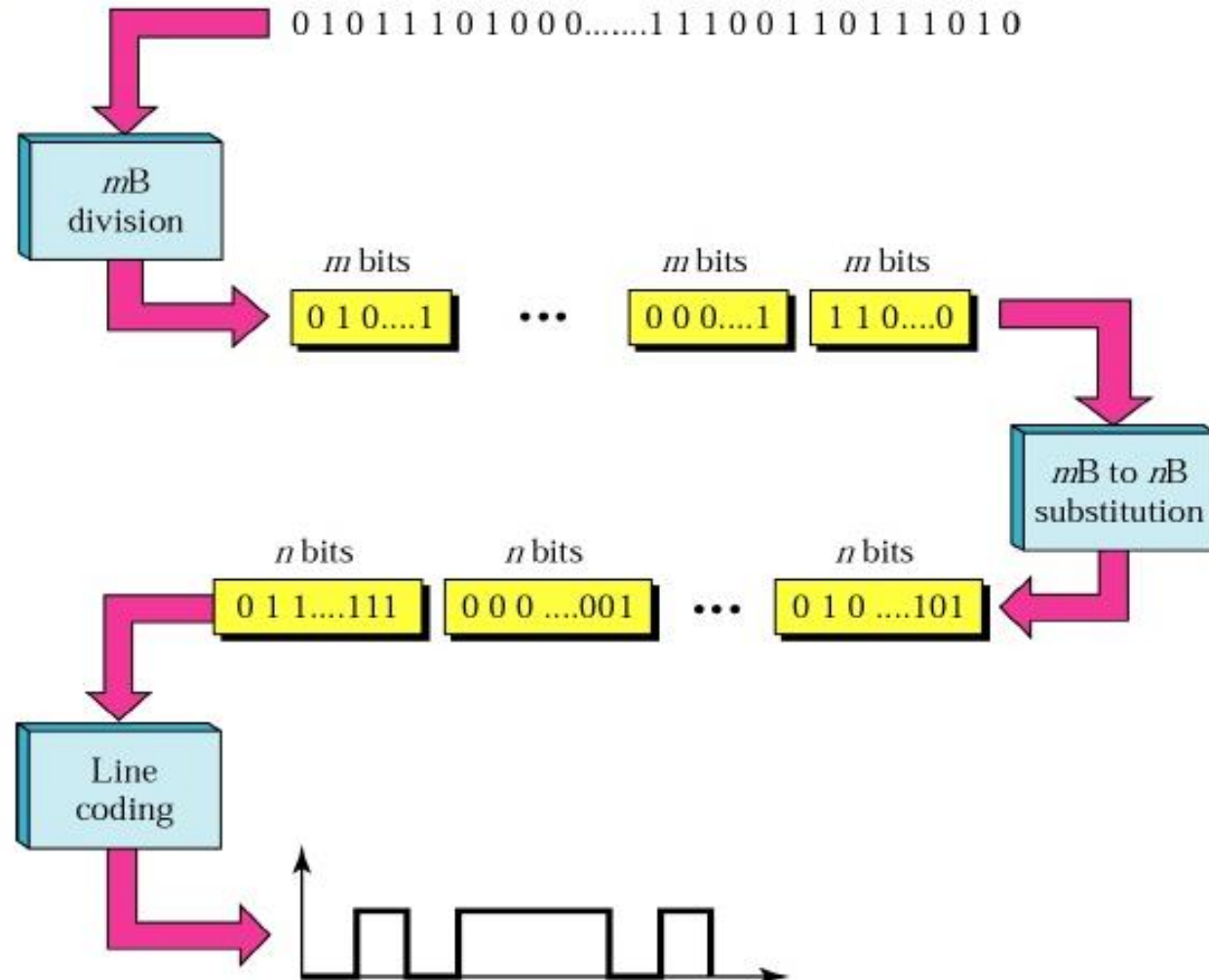


Figure 4.4 *Line coding schemes*



Coding

Block Coding



Coding

☑ Block Coding

- A 4 bit can have 16 combinations.
- A 5 bit can have 32 combinations.
- We therefore have $32 - 16 = 16$ extra groups that are not used for 4B/5B encoding.

Table 4.2 4B/5B mapping codes

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Coding

☑ Block Coding

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0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

- Some of the extra groups are used for control/signaling purposes.
- Error detection: If a 5-bit group arrives that belongs to the unused portion of the table, the receiver knows that there is an error in the transmission.

Channel Capacity

✓ **Channel Capacity:** Refers to the maximum data rate that can be reliably transmitted over a channel.

✓ **Signal-to-noise ratio (SNR, or S/N)**

$$(SNR)_{dB} = 10 \log_{10} \frac{\text{signal power}}{\text{noise power}}$$

✓ **Nyquist Theorem:** Defines channel capacity in a noiseless environment with B is the channel bandwidth (in Hz) and M is the number of signal levels.

$$\text{Nyquist data rate} = 2 \times B \times \log_2 M \text{ bps}$$

- A high SNR means a high-quality signal. SNR sets upper bound on achievable data rate

✓ **Shannon's Capacity Formula:** Provides the maximum achievable data rate considering both signal and noise.

$$\text{Shannon's capacity} = B \log_2(1 + S/N) \text{ bps}$$

Channel Capacity

✓ Examples

Nyquist data rate = $2 \times B \times \log_2 M$ bps

Assume that you have discovered a novel material that has negligible electrical noise. What is the maximum data rate that this material could achieve over a phone wire having a bandwidth of 3100 Hz if data was encoded with 64-QAM?

Solution:

We have $B = 3100$, $M = 64$

Data rate = $2 \times 3100 \times \log_2 64 = 37,200$ bps

Shannon's capacity = $B \log_2(1 + S/N)$ bps

For an SNR of 30dB, what is the maximum data rate that could be achieved over a phone wire having a bandwidth of 3100 Hz?

Solution :

$$10 \log_{10} S/N = 30$$

$$\log_{10} S/N = 3$$

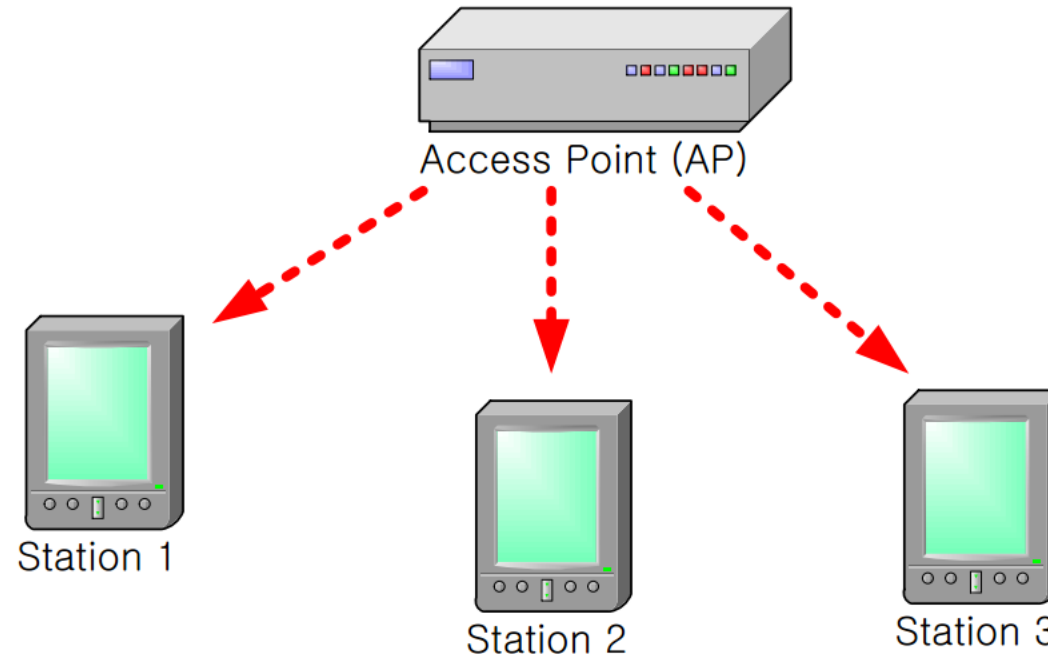
$$S/N = 10^3 = 1000$$

$$\text{Shannon's Capacity} = 3100 \log_2(1 + 1000) = 30,894 \text{ bps}$$

Wireless Access Channels

Broadcast Channel

- From AP to stations
 - i.e., **down**link(DL) or for**ward** link (FL)
- **Multiplexing** schemes used
 - e.g FDM, TDM, C DM, OFDM, WDM ...



Wireless Access Channels

☑ Duplexing Schemes

□ Frequency Division Duplexing (FDD)

- UL/DL in well separated frequency bands
- No synchronization issue
- Used for most commercial cellular systems

□ Time Division Duplexing (TDD)

- UL/DL shares the same frequency band in different time slots
- Need a guard time between UL and DL boundary
- Flexible UL/DL time sharing -> not really!
- Easy channel estimation due to channel reciprocity

Wireless Access Channels

☑ Transmission mode

Half-Duplex FDD

- Half-Duplex FDD is a communication mode where transmission and reception occur on separate frequencies but not simultaneously.

Example Use Cases

- 1. Walkie-Talkies (Two-Way Radios):** One frequency is used to talk, another to listen.
- 2. Satellite Communications:** Some satellites use half-duplex FDD for communication, alternating between sending and receiving signals.
- 3. Wireless Networks (Some LTE or IoT Applications):** Some wireless systems use half-duplex FDD to save power and reduce complexity.

Wireless Access Channels

☑ Transmission mode

Full Duplex

Full-Duplex FDD is a communication technique where transmission and reception occur **simultaneously** using **separate frequency channels**.

Comparison with Full-Duplex FDD

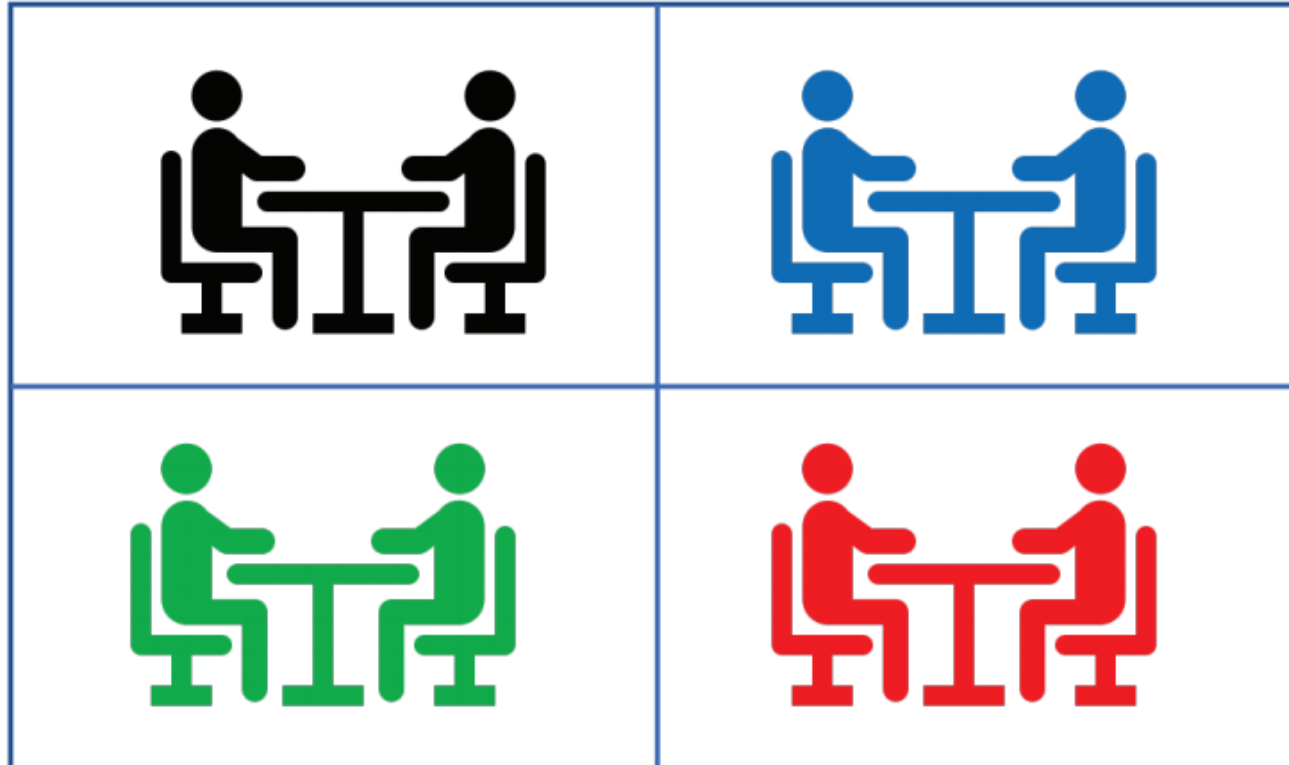
Feature	Half-Duplex FDD	Full-Duplex FDD
Frequency Usage	Two separate frequencies	Two separate frequencies
Simultaneous Tx/Rx	No	Yes
Complexity	Lower	Higher
Cost	Lower	Higher

Wireless Access Channels

✓ Multiple Access Methods

❖ Frequency Division Multiple Access (FDMA):

- ✓ Groups use different frequency bands simultaneously.
- ✓ Each group transmits on its allocated frequency band without overlapping with others.



(b) FDMA: Communicating groups are all talking at the same time, but in different rooms

Wireless Access Channels

✓ Multiple Access Methods

FDM/FDMA (Frequency Division)

- Channel spectrum divided into frequency bands
- Each station assigned fixed frequency band
- Unused transmission time in frequency bands go idle
- **Example: 6 stations, 1,3,4 have pkt, frequency band 2,5,6 idle**

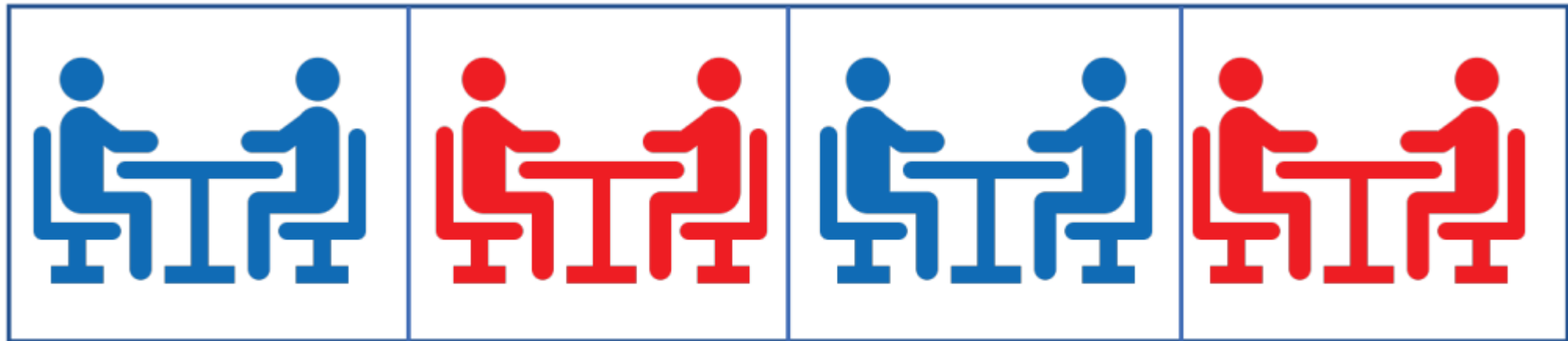


Wireless Access Channels

✓ Multiple Access Methods

Time Division Multiple Access (TDMA):

- Groups take turns using the same frequency band.
- Each group occupies the channel for a specific time slot, sequentially.



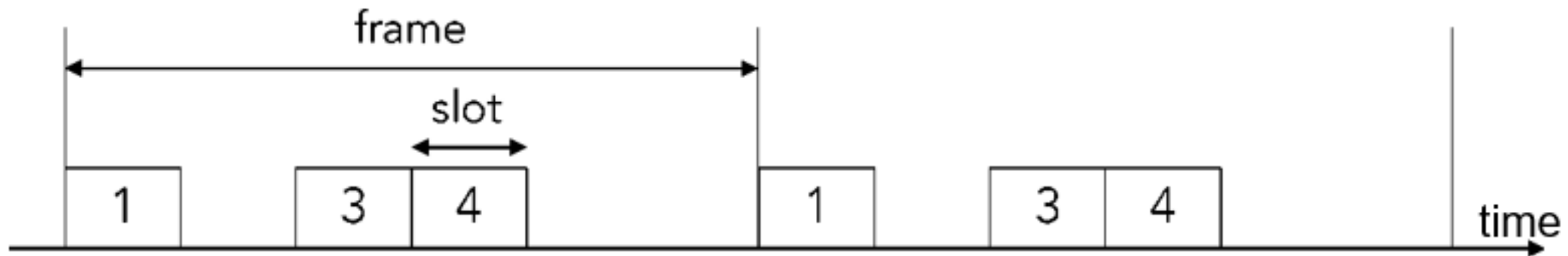
(a) TDMA: Communicating groups are taking turns

Wireless Access Channels

✓ Multiple Access Methods

TDM/TDMA (Time Division)

- Access to channel in "rounds"
- Each station gets fixed length slot (length = pkt trans time) in each round
- Unused slots go idle
- Example: 6 stations, 1,3,4 have pkt, slots 2,5,6 idle



Wireless Access Channels

✓ Multiple Access Methods

Code Division Multiple Access (CDMA):

- ✓ Groups transmit simultaneously on the same frequency band using different codes (sequences).
- ✓ Each group's signal is encoded with a unique code that allows receivers to distinguish between them despite overlapping transmissions.



(c) CDMA: Communicating groups are all talking at the same time in the same room

Wireless Access Channels

✓ Multiple Access Methods

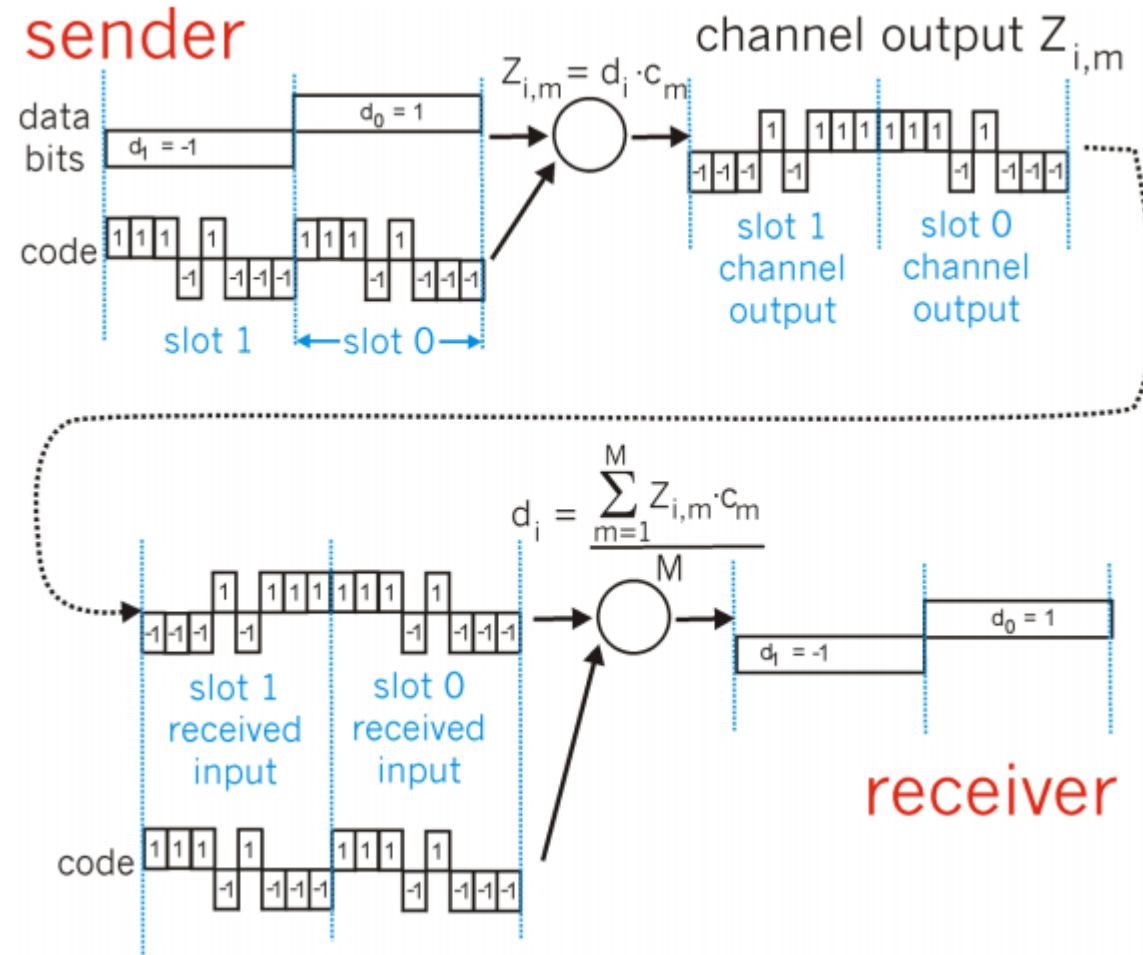
CDM/CDMA (Code Division)

- Unique "code" assigned to each user
- All users share same frequency, but each user has own "chipping" sequence (i.e., code) to encode data
- *encoded signal* = (original data) $\times$ (chipping sequence)
- *decoding*: inner-product of encoded signal and chipping sequence
- Allows multiple users to "coexist" and transmit simultaneously with minimal interference (if codes have good "cross-correlation")

Wireless Access Channels

✓ Multiple Access Methods

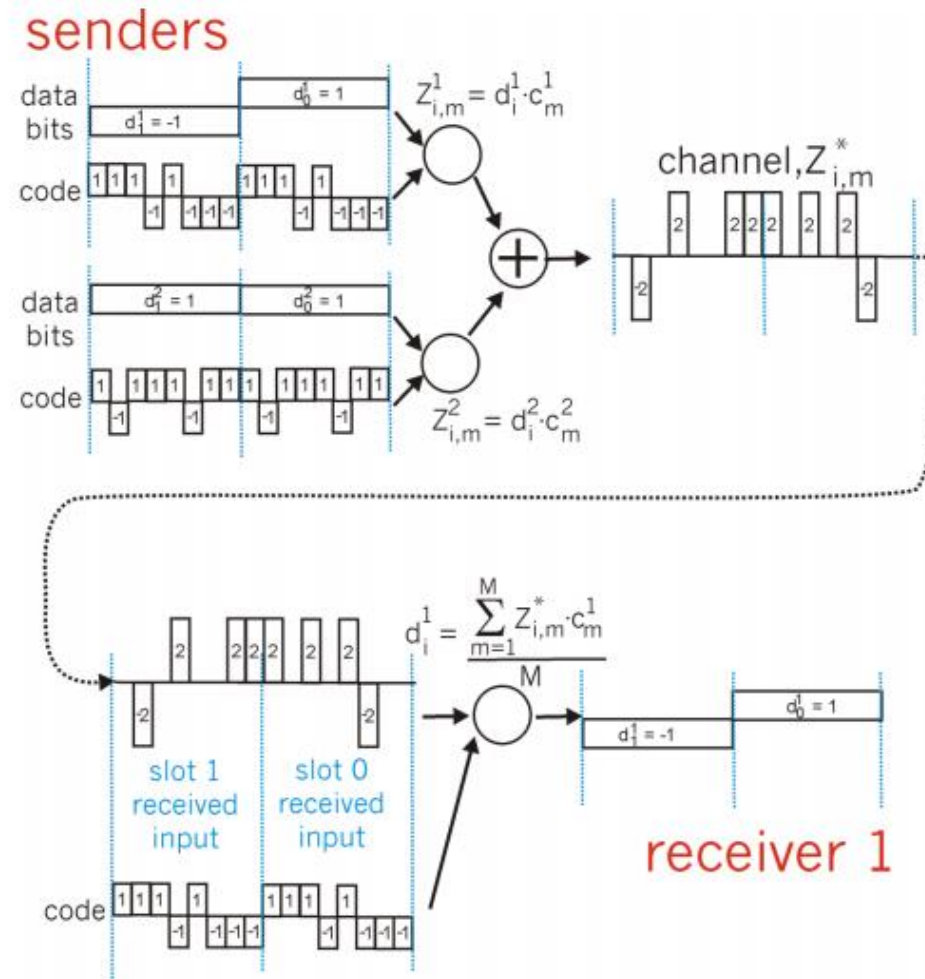
CDMA Encode/Decode



Wireless Access Channels

✓ Multiple Access Methods

CDMA: Two-sender Interference



Spread Spectrum

☑ Spread Spectrum

- ✓ Techniques that spread an **original narrow bandwidth** signal over a much **wider bandwidth**.
- ✓ Used for enhancing **security**, resistance to **interference**, **noise**, and **jamming**, and for implementing CDMA.

❖ Purposes of Spread Spectrum:

- ✓ **Security**: Makes the signal harder to detect with a narrowband receiver.
- ✓ **Interference Resistance**: Difficult to jam because the signal frequency changes rapidly.
- ✓ **CDMA (Code Division Multiple Access)**: Uses spreading codes as unique identifiers for different users.

Spread Spectrum

❖ Frequency Hopping Spread Spectrum (FHSS):

- ✓ Both transmitter and receiver use a synchronized sequence to switch frequencies rapidly within a band.
- ✓ Enhances security by changing frequencies unpredictably.
- ✓ Used in Bluetooth and military communications for anti-jamming and security.

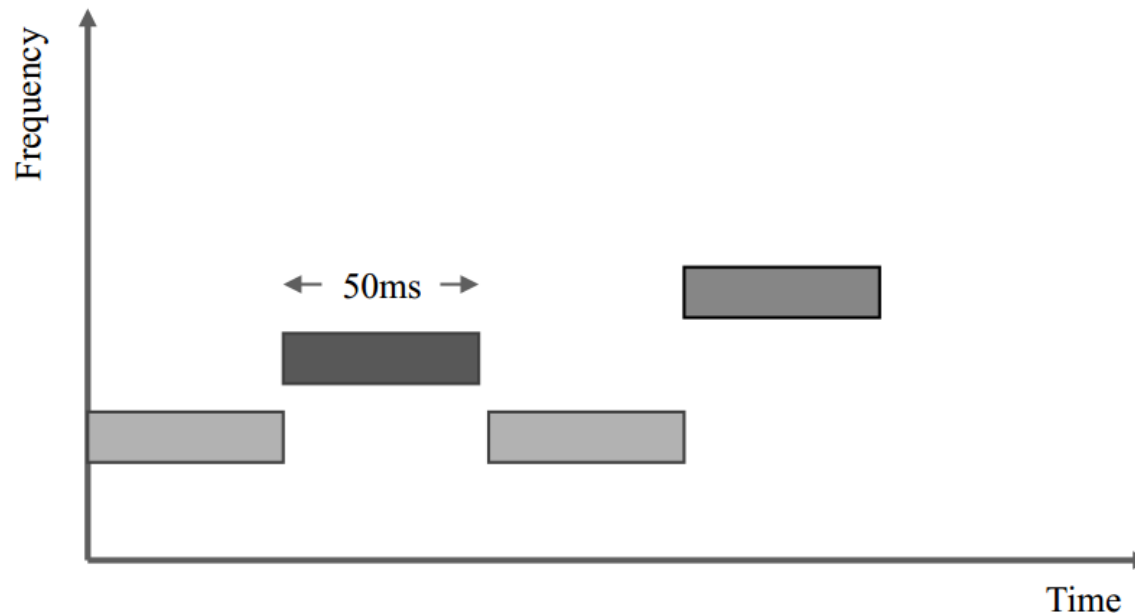


Fig. 11. Frequency hopping. The transmitter switches frequency every 50 ms.

Spread Spectrum

❖ Direct-Sequence Spread Spectrum (DSSS):

- ✓ Expands each data bit into a longer sequence (codeword) using a secret code known to the receiver.
- ✓ Increases signal bandwidth significantly compared to data bandwidth.
- ✓ Uses spreading factor to denote the number of code bits per data bit (e.g., 10 bits per data bit is common).

Spread Spectrum

❖ Direct-Sequence Spread Spectrum (DSSS)

User #1
Data : 0 0
Spreading code : 0 1 0 1
Data : 0 0 0 0 0 0 0 0
Spreading code : 0 1 0 1 0 1 0 1
X-OR : 0 1 0 1 0 1 0 1

User #2
Data : 1 0
Spreading code : 0 0 1 1
Data : 1 1 1 1 0 0 0 0
Spreading code : 0 0 1 1 0 0 1 1
X-OR : 1 1 0 0 0 0 1 1

User #3
Data : 1 1
Spreading code : 0 0 0 0
Data : 1 1 1 1 1 1 1 1
Spreading code : 0 0 0 0 0 0 0 0
X-OR : 1 1 1 1 1 1 1 1

All 3 users Tx at the same frequency and at the same time

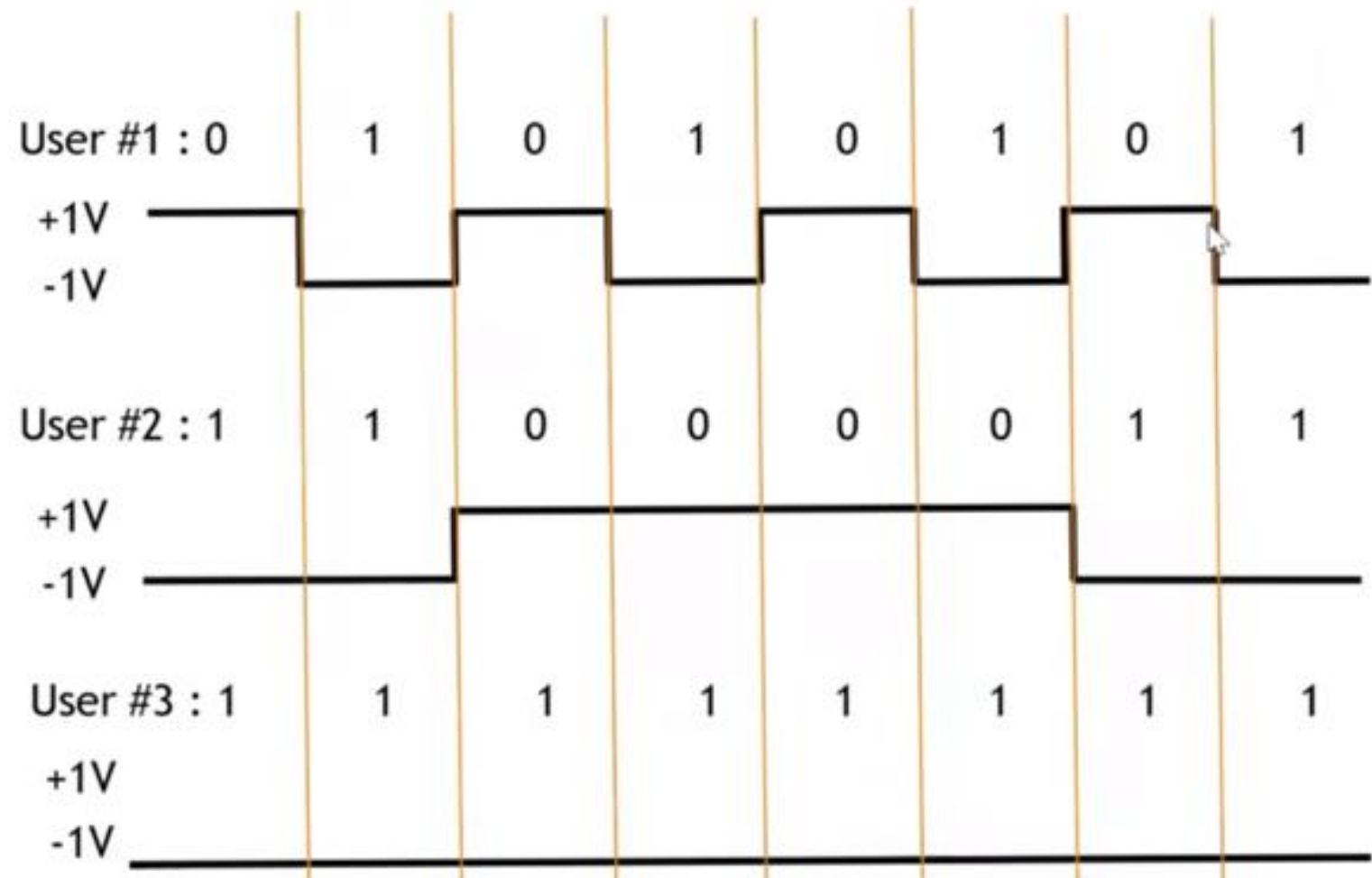
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

Spread Spectrum

❖ Direct-Sequence Spread Spectrum (DSSS)

Define: 0 → +1V

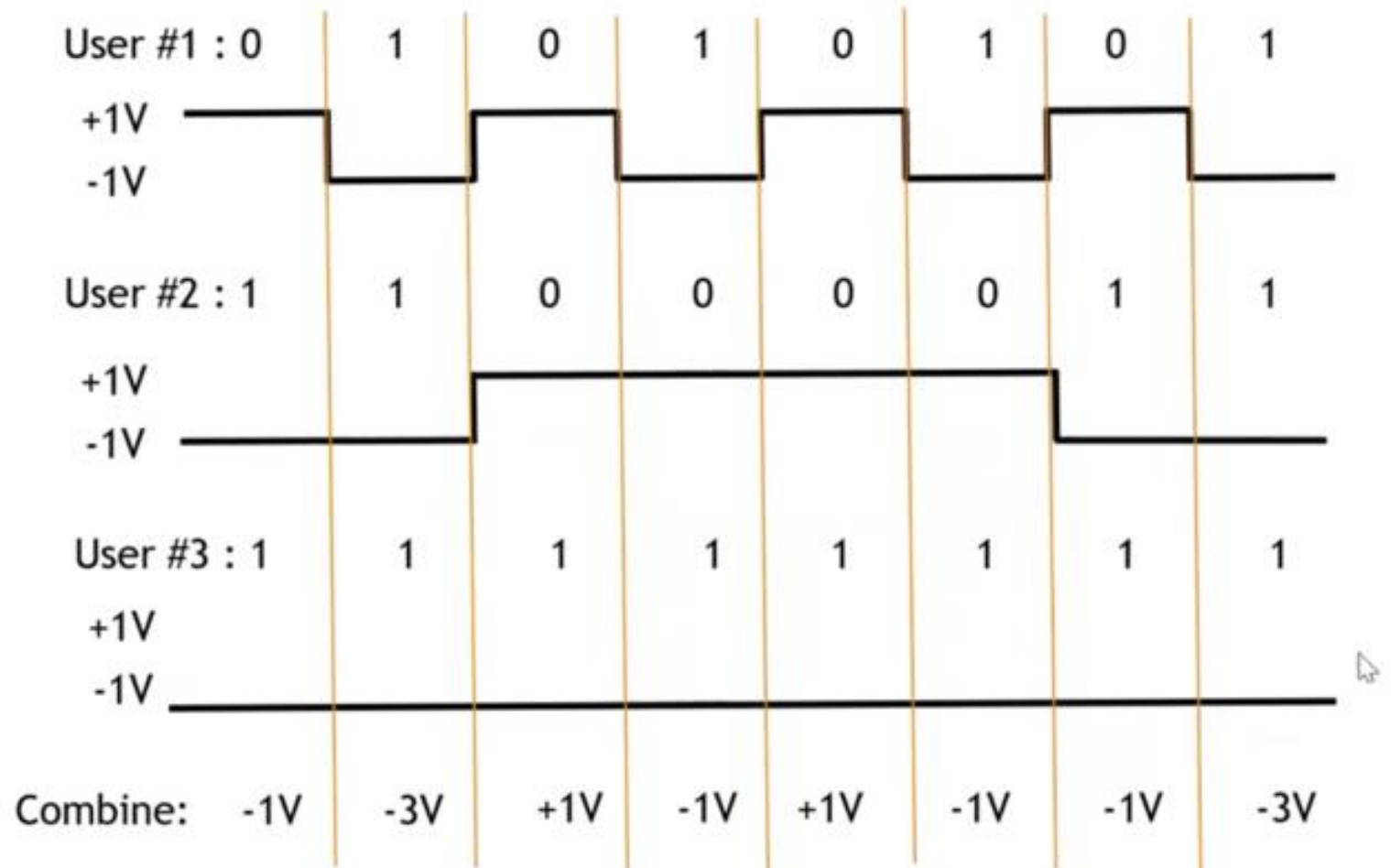
1 → -1V



Spread Spectrum

❖ Direct-Sequence Spread Spectrum (DSSS)

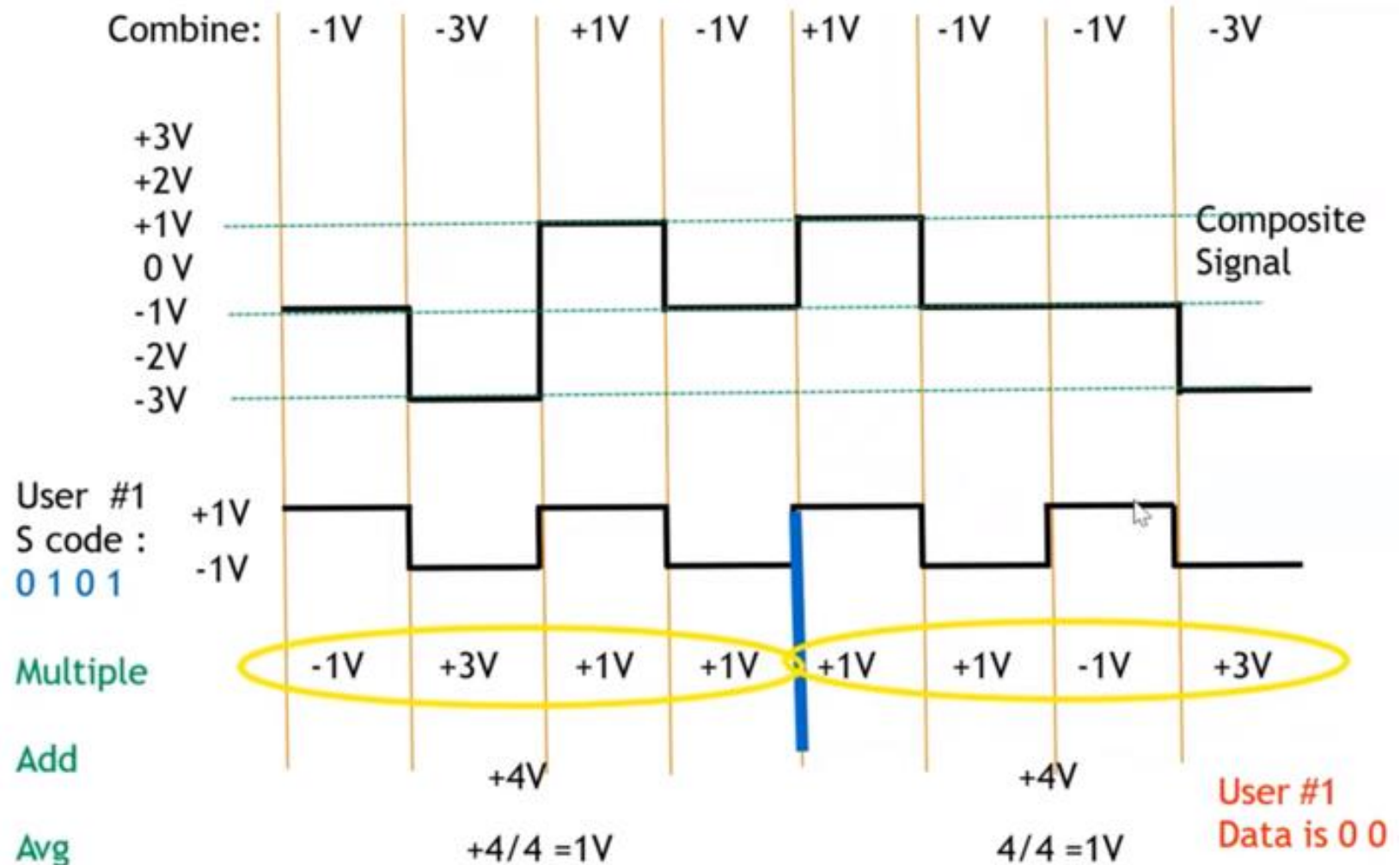
Combine all three signals



Spread Spectrum

❖ Direct-Sequence Spread Spectrum (DSSS)

- Recover user 1 data



THANKS

